

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Scala For Data Science

Practical VI

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Class: SYBsc IT	Subject: Scala For Data Science
Date of Assignment: 14/07/2026	Date/Time of Submission: 14/07/2026

AIM: Generate a random matrix using Breeze and compute its transpose and determinant.

CODE:

```
build.sbt  IQRA.scala x
1  import breeze.linalg._
2  import scala.util.Random
3
4  object Demo extends App {
5      val r = 3
6      val c = 3
7
8      val d = Array.fill(r * c)(Random.nextInt(10)).map(_.toDouble)
9      val m = new DenseMatrix(r, c, d)
10     val trans = m.t
11     val d1 = det(m)
12
13     println(s"Original Matrix: \n$m")
14     println(s"\nTranspoerted Matrix: \n$trans")
15     println(s"\nDeterminanat: $d1")
16     print("IQRA S021")
17 }
18
19
```

OUTPUT:

```
Transpoerted Matrix:
8.0  4.0  2.0
7.0  0.0  1.0
0.0  8.0  1.0

Determinanat: 19.999999999999996
IQRA S021
Process finished with exit code 0
```

Practical VII

AIM: Write a program to perform element-wise addition, subtraction, multiplication, and division of two Breeze matrices.

CODE:

```
build.sbt  IQRA.scala x
1  import breeze.linalg._
2  object Demo extends App{
3
4      val A = DenseMatrix((1.0, 2.0, 3.0),(4.0, 5.0, 6.0))
5      val B = DenseMatrix((6.0, 5.0, 4.0),(3.0, 2.0, 1.0))
6
7      val add = A + B
8      val sub = A - B
9      val mul = A *: B
10     val div = DenseMatrix.tabulate(A.rows, A.cols){ (i, j) => A(i, j) / B(i, j) }
11
12     println("Matrix A:")
13     println(A)
14     println("\nMatrix B:")
15     println(B)
16     println("\nAddition :")
17     println(add)
18     println("\nSubtraction :")
19     println(sub)
20     println("\nMultiplication :")
Demo
```

```
println("\nDivision :")
println(div)
println("\nIQRA S021")
```

OUTPUT:

```
"C:\Program Files\Java
```

```
Matrix A:
```

```
1.0  2.0  3.0
4.0  5.0  6.0
```

```
Matrix B:
```

```
6.0  5.0  4.0
3.0  2.0  1.0
```

```
Addition :
```

```
7.0  7.0  7.0
7.0  7.0  7.0
```

```
Subtraction :
```

```
-5.0 -3.0 -1.0
1.0  3.0  5.0
```

```
Multiplication :
```

```
6.0  10.0  12.0
12.0  10.0  6.0
```

```
Division :
```

```
0.16666666666666666  0.4  0.75
1.3333333333333333  2.5  6.0
```

```
IQRA S021
```

Practical VIII

AIM: Slice a Breeze matrix to extract a sub-matrix and calculate its row and column sums.

CODE:

```
build.sbt  IQRA.scala x
1  import breeze.linalg._
2  object Demo extends App {
3
4      val m = DenseMatrix(
5          (1.0, 2.0, 3.0, 4.0),
6          (5.0, 6.0, 7.0, 8.0),
7          (9.0, 10.0, 11.0, 12.0),
8          (13.0, 14.0, 15.0, 16.0)
9      )
10
11     val s = m(1 ≤ to ≤ 2, 1 ≤ to ≤ 2)    // 2x2 sub-matrix
12     val r = sum(s(*, ::))                // * means all rows, :: means row-wise
13     val c = sum(s(:, *))
14
15     println(s"Original Matrix:\n$m")
16     println(s"\nSub-Matrix : \n$s")
17     println(s"\nRow Sums: $r")
18     println(s"Column Sums: $c")
19     println("IQRA S021")
20 }
```

OUTPUT:

```
"C:\Program Files\Java\jdk1.8.0_
Original Matrix:
1.0  2.0  3.0  4.0
5.0  6.0  7.0  8.0
9.0  10.0 11.0 12.0
13.0 14.0 15.0 16.0

Sub-Matrix :
6.0  7.0
```

```

6.0    7.0
10.0   11.0

Row Sums: DenseVector(13.0, 21.0)
Column Sums: Transpose(DenseVector(16.0, 18.0))
IQRA S021

Process finished with exit code 0

```

Practical VIII

AIM: Find the correlation between two lists of numbers.
 Implement the formula for Pearson correlation coefficient.

CODE:

```

2      object Demo extends App {
20          denX = denX + (x(i) - meanX) * (x(i) - meanX)
21      }
22
23      for (i <- 0 until y.length) {
24          denY = denY + (y(i) - meanY) * (y(i) - meanY)
25      }
26
27      val den = math.sqrt(denX * denY)
28      val cor = num / den
29
30      println("Pearson Correlation Coefficient: " + cor)
31      println("IQRA S021")
32  }
33
34
35
36

```

OUTPUT:

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
```

```
Pearson Correlation Coefficient: 1.0
```

```
IQRA S021
```

```
Process finished with exit code 0
```